

Library Management System

Features 5 :

- Add book
- View book
- Search book
- Issue book
- Return Book

Member :

- Book Id
- Book Title
- Author Name
- Category
- Publisher
- Year
- Quantity
- Price

```
=====
LIBRARY MANAGEMENT SYSTEM
=====
```

1. Add Book
2. View Book
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Choice :

Functional Requirements :

1. Add Book :

Validate

- Duplicate ID
- Year between 1900-2026
- Quantity <=0
- Price<=0

output :

Added Books Successfully

2. View Book :

Display

ID

Title

Author Name

Publisher

Category
Year
Quantity
Price

3. Search Book :

Checks : The given Id

Returns “i” if it is found otherwise “-1”

4. Issue Book :

Quantity decreases after issuing the book

5. Return Book :

Quantity increases after returning the book

6. Exit.

IMPLEMENTATION

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
#define MAX 100
typedef struct
{
    int id;
    char title[40];
    char author[30];
    char category[35];
    char publisher[50];
    int year;
    int quantity;
    float price;
}library;

library books[MAX];
int count=0;
int findId(int id)
{
    for(int i=0;i<count;i++)
    {
        if(books[i].id==id)
            return i;
    }
    return -1;
}
int validyear(int y)
{
    if(y<1900 || y>2026){
        return 0;
    }
}
```

```

    return 1;
}

void addBook()
{
    library b;
    int id;
    printf("Enter Book id: ");
    scanf("%d",&b.id);
    if(findId(b.id) != -1)
    {
        printf("duplicate id\n");
        return;
    }
    getchar();
    printf("TITLE: ");
    fgets(b.title,40,stdin);
    b.title[strcspn(b.title,"\n")]=0;
    printf("AUTHOR NAME: ");
    fgets(b.author,30,stdin);
    b.author[strcspn(b.author,"\n")]=0;
    printf("CATAGORY: ");
    fgets(b.category,35,stdin);
    b.category[strcspn(b.category,"\n")]=0;
    printf("PUBLISHAR: ");
    fgets(b.publishar,50,stdin);
    b.publishar[strcspn(b.publishar,"\n")]=0;
    printf("PUBLISION YEAR: ");
    scanf("%d",&b.year);
    if(!validyear(b.year))
    {
        printf("INVALID YEAR.\n");
        return;
    }
    getchar();
    printf("QUANTITY: ");
    scanf("%d",&b.quantity);
    if(b.quantity<=0)
    {
        printf("Inavalid quantity\n");
        return;
    }
    getchar();
    printf("PRICE: ");
    scanf("%f",&b.price);
    if(b.price<=0)
    {
        printf("Invalid Price\n");
        return;
    }
    books[count++]=b;
    printf("\nAdd book succesfully.\n");
}

```

```

}

void viewBook()
{
    if(count==0)
    {
        printf("No books\n");
        return;
    }
    for(int i=0;i<count;i++)
    {
        library b = books[i];
        printf("BOOK ID:%d\n",b.id);
        printf("TITLE:%s\n",b.title);
        printf("AUTHOR NAME:%s\n",b.author);
        printf("CATEGORY:%s\n",b.category);
        printf("PUBLISHER:%s\n",b.publishar);
        printf("PUBLICATION YEAR:%d\n",b.year);
        printf("QUANTITY:%d\n",b.quantity);
        printf("PRICE:%.2f\n",b.price);
        printf("view succesusfully.\n");
    }
}

void searchBook()
{
    int id;
    library b;
    printf("Enter Book id: ");
    scanf("%d",&b.id);
    int po=findId(b.id);
    if(po == -1)
    {
        printf("the book is not found\n");
        return;
    }
    else
    {
        printf("-----BOOK DETAILS-----\n");
        printf("BOOK ID:%d\n",books[po].id);
        printf("TITLE:%s\n",books[po].title);
        printf("AUTHOR NAME:%s\n",books[po].author);
        printf("CATEGORY:%s\n",books[po].category);
        printf("PUBLISHER:%s\n",books[po].publishar);
        printf("PUBLICATION YEAR:%d\n",books[po].year);
        printf("QUANTITY:%d\n",books[po].quantity);
        printf("PRICE:%.2f\n",books[po].price);
        printf("find succesusfully.\n");
    }
}

```

```

void issueBook()
{
    library b;
    int id;
    printf("Enter ID: ");
    scanf("%d",&b.id);
    int po=findId(b.id);
    if(po == -1)
    {
        printf("The book is not found\n");
        return;
    }
    books[po].quantity--;
    printf("Book issued successfully.\n");
    printf("Remaining quantity:%d\n",books[po].quantity);
}

```

```

void returnBook()
{
    library b;
    int id;
    printf("Enter ID: ");
    scanf("%d",&b.id);
    int po=findId(b.id);
    if(po == -1)
    {
        printf("The book is not found\n");
        return;
    }
    books[po].quantity++;
    printf("The book return succesfully.\n");
    printf("Avalable quantity:%d\n",books[po].quantity);
}

```

```

int main()
{
    int ch;
    while (1)
    {
        printf("\n=====\\n");
        printf("    LIBRARY MANAGEMENT SYSTEM\\n");
        printf("=====\\n");
        printf("1. Add Book\\n");
        printf("2. View Books\\n");
        printf("3. Search Book\\n");
        printf("4. Issue Book\\n");
        printf("5. Return Book\\n");
        printf("6. Exit\\n");
        printf("=====\\n");
        printf("Enter your choice:");
        scanf("%d",&ch);
        switch (ch)

```

```
{
    case 1:
        addBook();
        break;
    case 2:
        viewBook();
        break;
    case 3:
        searchBook();
        break;
    case 4:
        issueBook();
        break;
    case 5:
        returnBook();
        break;
    case 6:
        printf("Thank you for using the Library Management System.\n");
        return 0;
    default:
        printf("Invalid choice.\n");
}
}
return 0;
}
```